

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence.

A compact citation keeps the observation traceable across agents and time. Encode in orbit, decode in AI’s reasoning with emem.

Jaya Kumari · Avijeet Singh · Vortex AI · avijeet@vortex.ai · whitepaper doi.org/10.5281/zenodo.20706317 · study doi.org/10.5281/zenodo.20706893
Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)

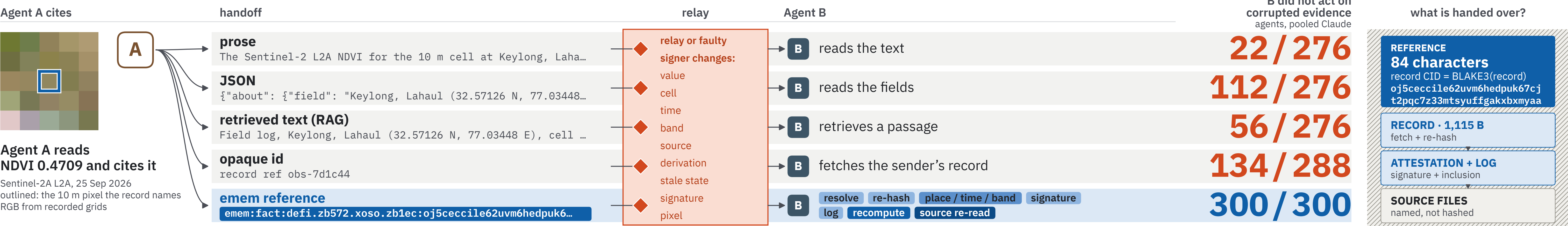
From observation to the next agent

Observe	Locate	Record	Hand off	Resolve	Check	Continue
Satellite or sensor Keep the source scene.	Location · product · time Lookup identity	Canonical record + CID Batch attestation	Pass the reference MCP · REST · A2A	Retrieve the same record Recover its provenance.	Hash · binding · signature Re-read source if needed.	Reason with cited evidence Carry the reference onward.

1 What survives an agent handoff?

The same observation, carried through receiver checks.

Agent A cites a source read. A relay changes the evidence. Agent B acts on it or checks the reference.



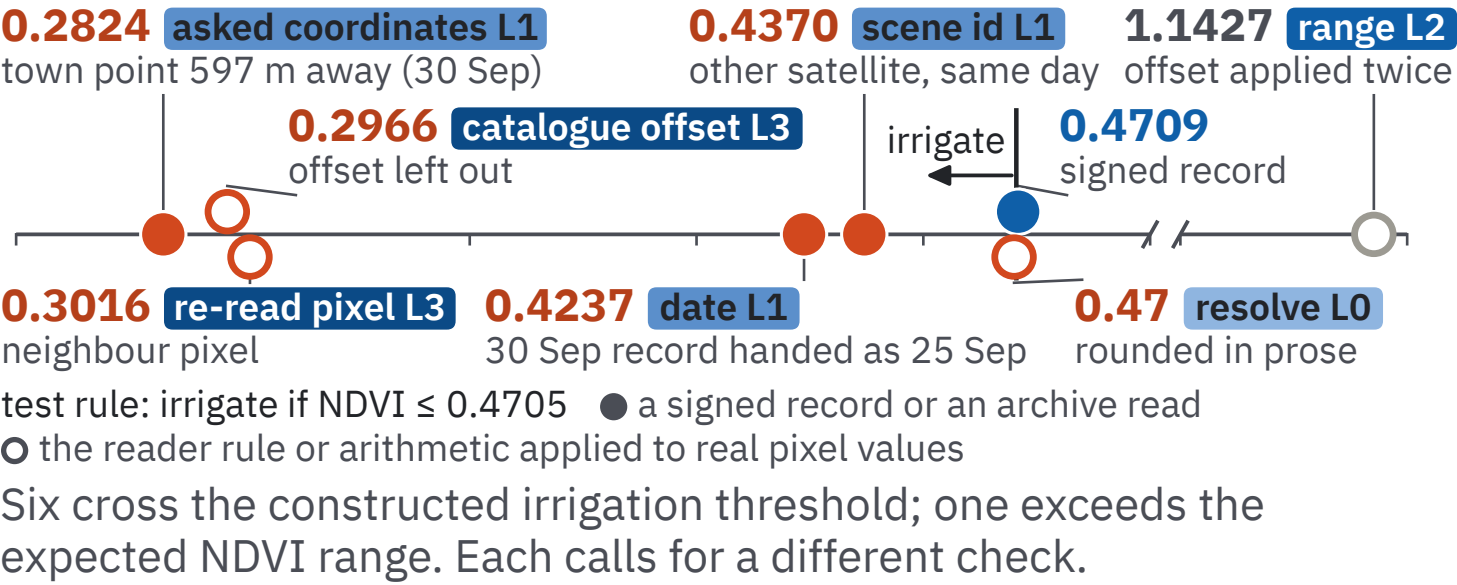
In the controlled handoff test, agents with a checked reference and explicit instructions did not act on corrupted evidence in 300 of 300 trials (264 declined, 36 used the genuine record); with prose, 22 of 276.

Earth data stays on the encoding device in orbit; its reference is downlinked and enters the AI’s reasoning (design; SAT-042 reference harness).

2 How does a value change between agents?

One NDVI, eight values

At Keylong on 25 Sep 2026, scene, date, pixel and arithmetic choices change the reported NDVI.



3 Where did the evidence change?

A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

CONCEPTUAL CHANGE MODEL

$$\Delta z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

Δz: change in readout. Environment · sensor · geolocation · encoder · residual. emem’s change attribution links evidence to each term.

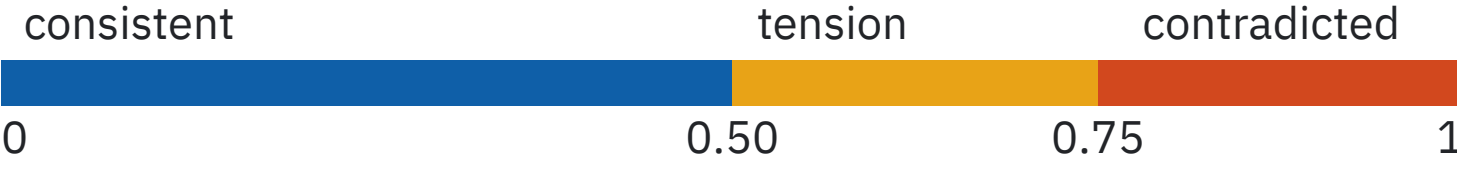
IMPLEMENTED · DRIFT-ANCHOR SCORE

$$d = |x - a| \text{ for finite } x, a \text{ and } 0 < \sigma < \infty$$

$$r = d / (3\sigma); s = r / (1 + r)$$

Invalid op: s = 0 if d = 0; otherwise s = 1.

x: output · a: anchor · σ: anchor uncertainty



Tests pin 3σ at 0.50 and 9σ at 0.75.

Score boundaries: consistent < 0.50; tension < 0.75.

The implemented score measures disagreement with an anchor. SAT-042 applies it to a fixed reference in the execution harness.

7 What does a checked reference establish?

Know what was checked

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Derived derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

FULL TRACE · 30 SEP 2026

Token record source pixel

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG(row, col) = DN(b) & Ed25519(o) & leaf in STH & STHgeo.qa prefix of STH

Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked 15 links (17 checks) in 17.9 s, all verified. Grey: what must still be trusted after each check.

1-3 content address 1,115 B hash to the fact cid; one flipped mantissa bit gives mu5x2cks... · still trusted: nothing

4-5 cell binding hashed cell = token cell = the lat/long the derivation used · nothing

6-7 derivation scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem’s code

8-9 upstream the STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum

10-11 server signature emem.dev’s Ed25519 receipt over the request id, time, cells and returned fact_cids · shows what emem.dev served, not that the value is right

12-14 public log entry 2,457,078 is in the 2,556,451-entry log, consistent with the root geo.qa co-signed · geo.qa is also run by Vortex AI

15 signing key the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC

Links 1-5 and 12-14 fit in a 4,906-byte file that repeats 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

THE CONTRIBUTION

Checkable observation handoff

An agent passes a reference to a specific physical observation; the receiver resolves it and checks the evidence.

Built with standard BLAKE3, Ed25519, CBOR and Merkle logs. STAC, openEO, PROV, C2PA, RAG and temporal storage provide complementary layers.

12 Use emem. Carry the evidence forward.

Measured separately: same record and value through 11 client paths; receipt signatures checked on 9 (30 Sep 2026).

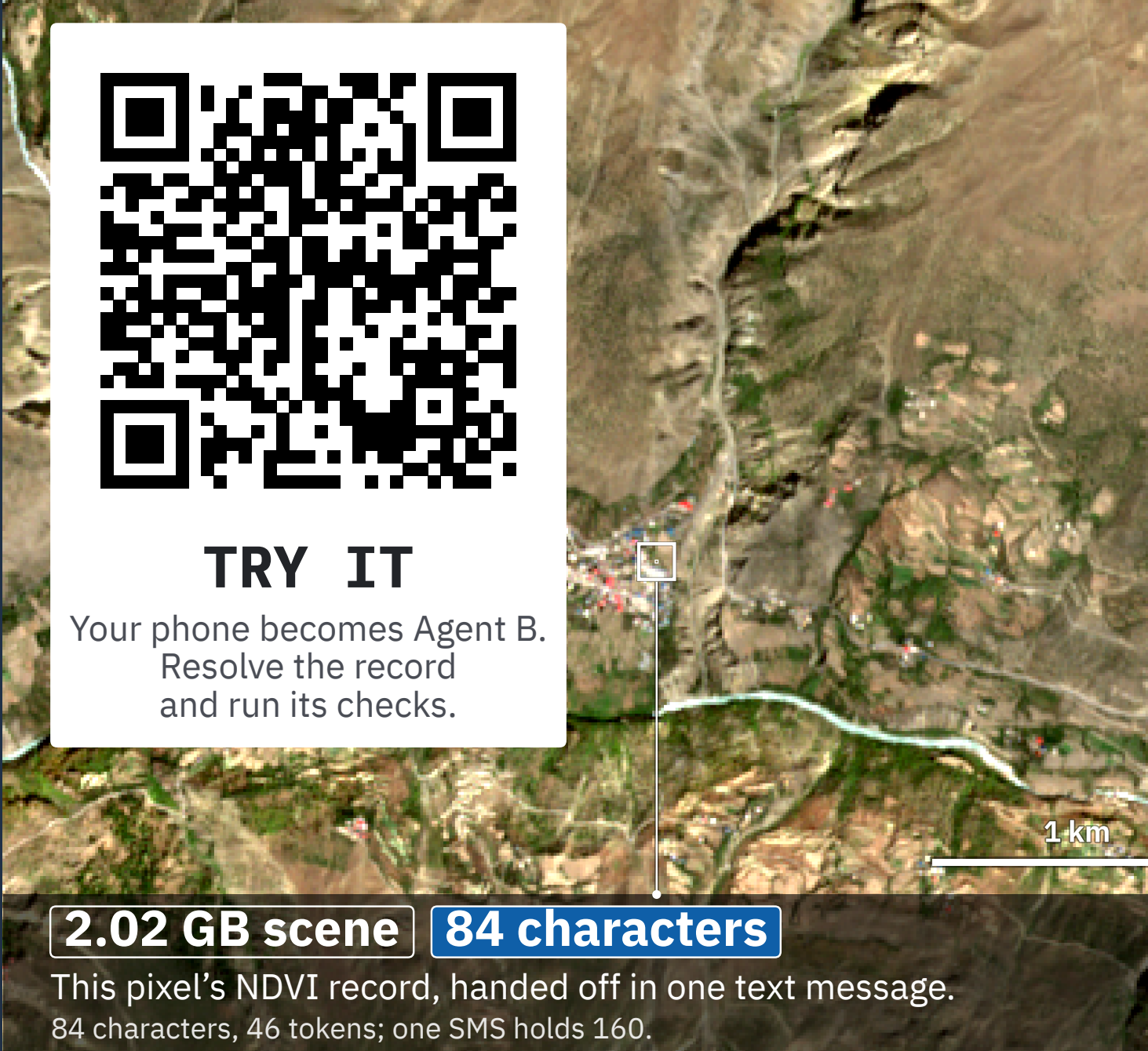
ChatGPT · @emem plugin “Signed facts about real places” Enable emem; paste a token.	Claude · Directory plugin / MCP “hand another agent a token it can check offline” Resolve from chat or terminal.	Dify · Marketplace plugin “Shared memory for AI agents. One address per fact, one signature you check.” Referent Lock / EUDR workflows	GitHub MCP Registry · Server listing “Satellites for AI.” Install MCP server	ClawHub · OpenClaw plugin listing “Shared, signed and content-addressed memory of the physical world...” Install the OpenClaw plugin.
---	--	--	--	---

Implementation: emem 8e9b401. Measurements: 29 Sep to 1 Oct 2026 · The linked methods connect this poster’s figures to their records, source data and reproducible checks. · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066) · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC V1.13 · JRC GFC2020 V3/4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)

Agent A cites a satellite observation. What can agent B check without trusting A, A’s model, or us?

emem makes raw satellite observations addressable by place, band and time, each a signed record named by the BLAKE3 hash of its bytes. The exact evidence A cites survives compaction, a change of model and a handoff; B re-hashes it, verifies the log entry and receipt offline, and traces it to the source pixel. Agents cannot write observations, and a changed, rounded or forged value no longer matches its name.

Trace (token) · record bytes · cell · pixel · receipt · log · key



2.02 GB scene 84 characters

This pixel’s NDVI record, handed off in one text message.

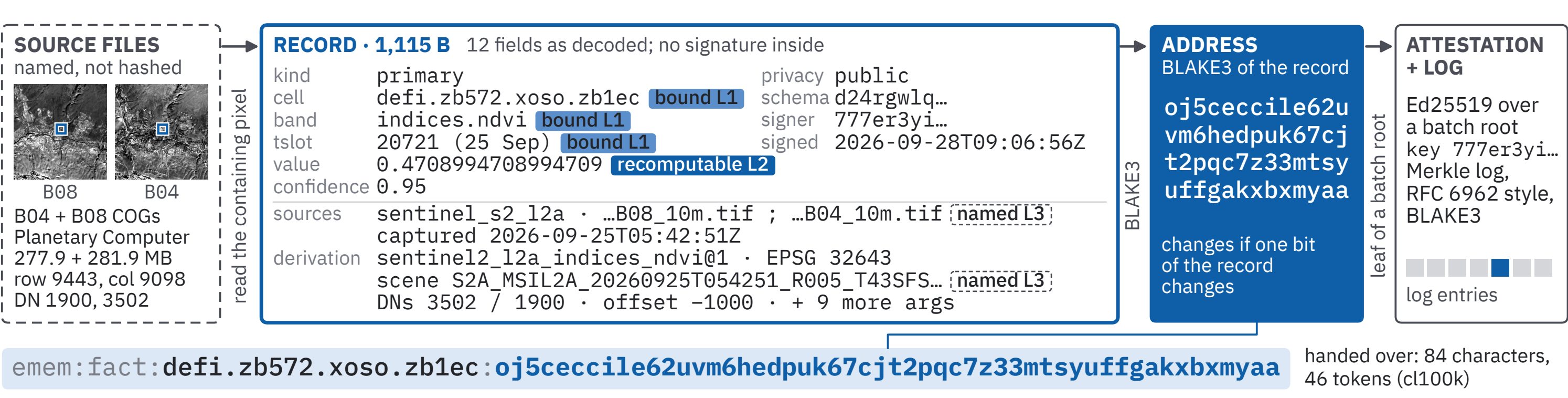
84 characters, 46 tokens; one SMS holds 160.

4 What exactly is handed over?

A compact reference to an exact record

The receiver uses an 84-character reference to fetch the record and its evidence.

INSPECT THE RECORD The 1,115 bytes behind the main example, decoded.



BLAKE3 identifies the 1,115-byte record. Its fields name the source files, sampled point and derivation; a batch signature covers the record address. Source-file hashes are absent from this record.

THE MEMORY, AS CODED · emem 18adb67 · fact.rs, cbor.rs, geo.rs, recall.rs, attest.rs, translog.rs, emem-attest

fact f = (kind, cell, band, tslot, value, unit, confidence, sources, derivation, fn key, args, privacy, schema_cid, signer, signed_at)
name b = CBOR(f), declaration order, NaN as f97600, -0 as +0; fact_cid = base32(BLAKE3-256(b)); token emem:fact:cell:fact_cid
cell lat_q = round((lat+90)/180*(2²¹-1)), lng_q = round((lng+180)/360*(2²¹-1)); pitch 8.583e-5° = 9.52 × 8.06 m here
recall tslot = floor(unix / 86 400) (20721 = 25 Sep 2026); key (cell, band, tslot); as of t: argmax (signed_at, cid) over signed_at ≤ t
sign R = MexKle(sorted fact_cids); attestation = Ed25519(PreimageV1(“attestation”){R, registry, schema}); a fact carries no signature
log leaf = BLAKE3(0x00 || BLAKE3(CBOR(attestation))), node = BLAKE3(0x01 || 1 || i); RFC 6962 shape
receipt sig = Ed25519(BLAKE3(“emem.preimage.v10” || request_id || served_at || primitive || cells || fact_cids || merkle))

5 Which changes can the receiver detect?

Checks can continue back to the source

Resolve Re-hash Bind Recompute Re-read

agents as B: three Claude models pooled, 1 Oct 2026

	prose	JSON	RAG	opaque id	emem, full check	first check that refuses
B acted on corrupted evidence	unaffected	refused	not applicable			
M2 seen in production	the only check that stops it	refused	flips the decision			
value	nothing altered: accepted everywhere, never refused	0/12	0/12	0/12	3/12	C L0
M1	stated value moved by 1 ULP	6/12	6/12	1/12	12/12	C L0
M2	stated value rounded to “0.47”	0/12	6/12	0/12	6/12	F L0
M8	value 0.45, re-encoded and re-hashed	0/12	6/12	0/12	6/12	F L0
cell	record cited for another cell	5/12	12/12	6/12	6/12	E L1
M9	cell changed inside the record, re-hashed	5/12	1/12	6/12	6/12	F L0
time	tslot changed to look current, re-hashed	0/12	6/12	6/12	6/12	F L0
M10	a record for another band handed over	0/12	12/12	6/12	10/12	E L1
band	source scene id changed, re-hashed	0/12	0/12	0/12	0/12	F L0
M11	offset changed, value recomputed, re-hashed	0/12	1/12	0/12	4/12	F L0
derivation	value (0.46) disagrees with the signed DNs	0/12	6/12	0/12	6/12	H L2
M14	an older record handed over as current	0/12	11/12	6/12	11/12	E L1
M5	a second signed version shown only to B	0/12	6/12	6/12	6/12	G L0
M16	1 ULP changed inside the served bytes	0/12	0/12	0/12	0/12	D L0
signature / id	token miscopied by one character	n/a	n/a	n/a	12/12	C L0
M3	value 0.45 under the forger’s own key	0/12	6/12	0/12	6/12	F L0
M13	DNs read from the pixel 10 m south	0/12	0/12	0/12	0/12	I L3
source pixel	same record; A meant another entity (out of scope)					
M15						
entity						
M17						

B did not act on corrupted evidence

22/276 112/276 56/276 134/288 300/300
R1 0/15 R1 0/15 R13/16 R116/16

C resolve L0 D hash L0 E binding L1 F signature L0 G log L0 H recompute L2 I re-read L3 D to I are cumulative: each adds one check

Bars: B did not act on it (declined, or used the genuine value), n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5 runs); squares: deterministic receiver. Totals pool 23 items, 1 Oct 2026.

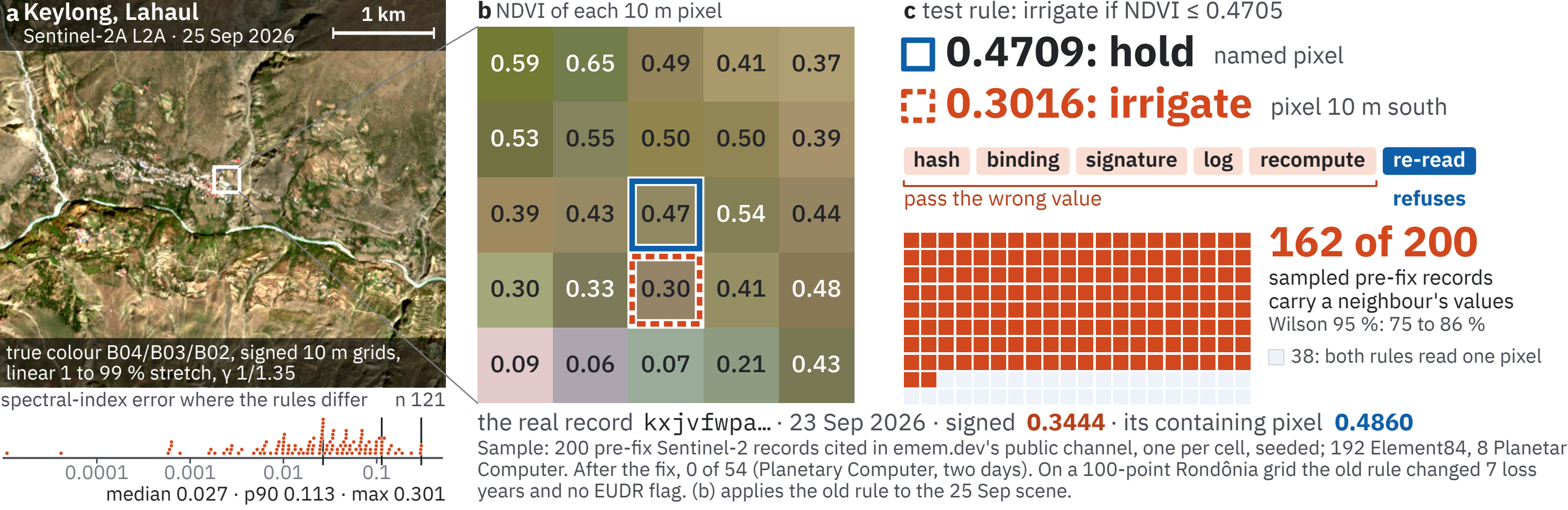
Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

Deeper checks expose different corruptions. Checked-reference agents made the expected decision on 71 of 72 genuine controls; one was refused (pooled Claude).

Agent bars: 12 runs per item, across 23 items. JSON and an opaque id caught some changes from their fields. In the live pre-fix record case, agents acted on 4 of 4 untrusted, 2 of 4 with resolve-and-bind instructions, and 0 of 4 with a source re-read.

6 A valid record can preserve a wrong source read

NDVI = (DN₈ − DN₄) / (DN₈ + DN₄ + 2o), o = −1000 (product offset)



CONNECT



Setup · EO tasks
Reproduce